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Chiral Superconductivity in Rhombohedral Graphene

Abstract

We integrate the discovery of chiral superconductivity in rhombohedral graphene (arXiv:2408.15233, 2024) into the UQFF framework. The chiral pairing gap function:

$$\Delta_{\text{chiral}}(\mathbf{k}) = \Delta_0 \cdot (k_x \pm i k_y)^d \cdot \exp\left(-\frac{[\text{SSq}] \cdot 10}{26}\right)$$

is modelled as $[\text{SCm}]$ pairing at quantum level $n = 10$, where d is the chiral winding number. The critical temperature:

$$T_c = \frac{\hbar \omega_D}{k_B} \cdot \exp\left(-\frac{1}{N(0) \cdot V_{\text{SCm}}}\right)$$

connects the graphene phonon Debye frequency ω_D to the $[\text{SCm}]$ pairing potential V_{SCm}. This establishes rhombohedral graphene as a condensed-matter analogue of the cosmic $[\text{SCm}]$ superconductive vacuum.

1. Introduction

Rhombohedral (ABC-stacked) multilayer graphene has emerged as a platform for exotic quantum states, including superconductivity with broken time-reversal symmetry — chiral superconductivity. The 2024 discovery (arXiv:2408.15233) reported chiral pairing consistent with topological $p + ip$ or $d + id$ order parameters.

Within UQFF, superconductivity at any scale is a manifestation of the $[\text{SCm}]$ vacuum condensate. Rhombohedral graphene at level 10 provides a laboratory-accessible probe of the same pairing mechanism that operates cosmologically.

2. Chiral Gap Function

2.1 [SCm] Pairing Model

The chiral gap function in UQFF notation:

$$\Delta_{\text{chiral}}(\mathbf{k}) = \Delta_0 \cdot |\mathbf{k}|^d \cdot \exp\left(-\frac{[\text{SSq}] \cdot n}{26}\right)$$

Parameter	Value	Description
Δ_0	10^{-4} eV	Bare pairing gap
d	2	Chiral winding number (d-wave)
n	10	UQFF quantum level
[SSq]	0.57	Squeeze-state parameter
$\exp(-[\text{SSq}] \cdot 10/26)$	0.8029	Level-10 suppression

2.2 Winding Number Interpretation

The chiral winding number d determines the topology: - $d = 1$: $p + ip$ pairing (single vortex) - $d = 2$: $d + id$ pairing (double vortex) - $d = 3$: $f + if$ pairing (higher angular momentum)

Each corresponds to a different [SCm] angular momentum sector in the 26D hierarchy.

3. Critical Temperature

3.1 BCS-Like Formula with [SCm] Potential

$$T_c = \frac{\hbar\omega_D}{k_B} \cdot \exp\left(-\frac{1}{N(0) \cdot V_{\text{SCm}}}\right)$$

Parameter	Value	Description
$\hbar$	1.055×10^{-34} J·s	Reduced Planck constant
ω_D	2×10^{13} rad/s	Graphene Debye frequency
k_B	1.381×10^{-23} J/K	Boltzmann constant
$N(0) \cdot V_{\text{SCm}}$	0.3	Coupling product

For $N(0) \cdot V_{\text{SCm}} = 0.3$: $T_c \approx 5.6$ K.

3.2 N(0)·V_SCm Sweep

$N(0) \cdot V_{\text{SCm}}$	T_c (K)	Regime
0.1	6.9×10^{-5}	Weak coupling
0.2	0.105	Intermediate
0.3	5.57	Moderate
0.5	20.7	Strong coupling
0.8	43.6	Very strong

4. Condensation Energy

The condensation energy per unit cell:

$$E_{\text{cond}} = \frac{1}{2}\Delta_0^2 \cdot \exp\left(-\frac{[\text{SSq}] \cdot 10}{26}\right)$$

provides the energy scale for the transition from normal to [SCm]-paired state.

5. Conclusions

Rhombohedral graphene chiral superconductivity provides a condensed-matter analogue of cosmic [SCm] pairing. The level-10 assignment in the UQFF hierarchy connects laboratory observations to the 26D compressed gravity framework. CP4 class `ChiralSCmGraphenePairingCalculator` (#619) implements gap function, T_c, and winding number computations.

References

1. arXiv:2408.15233 (2024)
2. Murphy, D.T., Star-Magic UQFF Framework (2024–2026)